



University Examination Previous Papers

Numerical method and analysis (SRM Institute of Science and Technology)



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B.Tech. / M.Tech (Integrated) DEGREE EXAMINATION, JULY 2023
Third Semester

21MAB206T – NUMERICAL METHODS AND ANALYSIS
(For the candidates admitted from the academic year 2022-2023)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

PART – A (20 × 1 = 20Marks)

Marks BL CO PO

Answer ALL Questions

- | | | | | |
|---|---|---|---|---|
| 1. The condition for convergence for iterative methods of solving simultaneous linear equations states that the coefficient matrix is | 1 | 1 | 1 | 1 |
| (A) Upper triangular | | | | |
| (B) Diagonally dominant | | | | |
| (C) Rectangular | | | | |
| (D) Singular | | | | |
| 2. The positive root of the equation $e^x = 2 + x$ lies between | 1 | 1 | 1 | 1 |
| (A) 1.1 and 1.2 | | | | |
| (B) 1.0 and 1.1 | | | | |
| (C) 1.2 and 1.3 | | | | |
| (D) 1.3 and 1.4 | | | | |
| 3. The order of convergence of N-R method is | 1 | 1 | 1 | 1 |
| (A) 1 | | | | |
| (B) 2 | | | | |
| (C) 3 | | | | |
| (D) 4 | | | | |
| 4. In Gauss elimination method to solve $Ax=B$, we use | 1 | 1 | 1 | 1 |
| (A) Forward substitution method | | | | |
| (B) Indirect substitution method | | | | |
| (C) Direct substitution method | | | | |
| (D) Back substitution method | | | | |
| 5. The relation between E and ∇ is | 1 | 2 | 2 | 1 |
| (A) $\nabla - E^{-1} = 1$ | | | | |
| (B) $1 + \nabla = E^{-1}$ | | | | |
| (C) $\nabla = 1 - E^{-1}$ | | | | |
| (D) $\nabla = 1 + E^{-1}$ | | | | |
| 6. $\Delta \tan^{-1} x =$ | 1 | 2 | 2 | 2 |
| (A) $\tan^{-1}(x+h) - \tan^{-1} x$ | | | | |
| (B) $\tan^{-1}(x+h) + \tan^{-1} x$ | | | | |
| (C) $\tan^{-1} x + \tan^{-1}(x+h)$ | | | | |
| (D) $\tan^{-1} x - \tan^{-1}(x+h)$ | | | | |
| 7. When the values of the independent variable are not equally spaced, then we apply | 1 | 2 | 2 | 1 |
| (A) Central difference formula | | | | |
| (B) Newton's forward formula | | | | |
| (C) Newton's backward formula | | | | |
| (D) Lagrange's formula | | | | |

8. The n^{th} forward difference of y are denoted by

1 1 2 1

- (A) $\Delta y_{n-1} = y_n - y_{n-1}$ (B) $\Delta y_n = y_{n-1} + y_{n+1}$
 (C) $\Delta y_{n-1} = y_n + y_{n-1}$ (D) $\Delta y_n = y_{n-1} + y_n$

9. The trapezoidal rule is

1 1 3 2

- (A) $\frac{h}{2}[(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})]$
 (B) $h[(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})]$
 (C) $h\left[\left(\frac{y_0 + y_n}{2}\right) + 2(y_1 + y_2 + \dots + y_{n-1})\right]$
 (D) $\frac{h}{2}[(y_0 + y_n) + 2(y_1 + y_3 + \dots) + 4(y_2 + y_4 + \dots)]$

10. Simpson's 3/8 rule is applied when

1 1 3 1

- (A) Number of ordinates is odd (B) Number of ordinates is even
 (C) n is multiple of three (D) n is even

11. Newton's forward formula to get the first derivative at $x=x_0$ is

1 1 3 1

- (A) $y' = \frac{1}{h} \left[\Delta^2 y_0 - \frac{\Delta^3 y_0}{2} + \frac{\Delta^5 y_0}{3} - \dots \right]$
 (B) $y' = \frac{1}{h} \left[\Delta y_0 - \frac{\Delta^2 y_0}{2} + \frac{\Delta^3 y_0}{3} - \dots \right]$
 (C) $y' = \frac{1}{h} \left[\Delta y_0 + \frac{\Delta^2 y_0}{2} + \dots \right]$
 (D) $y' = \frac{1}{h^2} \left[\Delta^2 y_0 - \frac{\Delta^3 y_0}{3} + \frac{\Delta^5 y_0}{3} + \dots \right]$

12. The value of $\sec 31^\circ$ from the following table is given by

1 1 3 2

θ°	31	32	33
$\tan \theta$	0.6008	0.6249	0.6494

- (A) 1.3835 (B) 1.1702
 (C) 1.6643 (D) 1.2901

13. Taylor's series method is

1 2 4 1

- (A) Single step method (B) Multi step method
 (C) Iterative method (D) Trial and error method

14. How many prior values require to predict the next value in Milne's method?

1 1 4 1

- (A) 4 (B) 2
 (C) 3 (D) 1

15. The modified Euler's method is based on the averages of

1 1 4 1

- (A) Points (B) Slopes
 (C) Curves (D) Both points and slopes

16. In R-K method Δy stands for 1 1 4 1
- (A) $\frac{1}{6}(k_1 + k_2 + k_3 + k_4)$ (B) $\frac{1}{6}(k_1 - 2k_2 + 2k_3 - k_4)$
- (C) $\frac{1}{6}(2k_1 + k_2 + k_3 + 2k_4)$ (D) $\frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$
17. If $B^2 - 4AC < 0$ then the second order PDE is known as 1 1 5 1
- (A) Elliptic (B) Parabolic
- (C) Hyperbolic (D) Poisson
18. The equation $\nabla^2 u = f(x, y)$ is called 1 1 5 1
- (A) Laplace equation (B) One dimensional heat equation
- (C) One dimensional wave equation (D) Poisson equation
19. $u_{ij} = \frac{1}{4}[u_{i-1,j-1} + u_{i+1,j+1} + u_{i+1,j-1} + u_{i-1,j+1}]$ 1 1 5 1
- (A) Standard five point formula (B) Diagonal five point formula
- (C) Explicit formula (D) Implicit formula
20. In crank Nicholson method the value of k for $\lambda=1$ is 1 1 5 1
- (A) $k = ah^2$ (B) $k = \frac{a}{2}h^2$
- (C) $k = ah$ (D) $k = h^2$

PART – B (5 × 8 = 40 Marks)

Answer ALL Questions

21. a. Find a solution of $3x + \sin x - e^x = 0$ correct to 4 decimal places by Newton's method. Marks 8 BL 3 CO 1 PO 2

(OR)

- b. Solve $e^x - 3x = 0$ by the method of iteration. 8 3 1 2
22. a. Using Newton's forward interpolation formula find the cubic polynomial which takes the following values. 8 3 2 2

$x:$	0	1	2	3
$f(x):$	1	2	1	10

(OR)

- b. Use Lagrange's interpolation formula to fit a polynomial to the data. 8 3 2 2

$x:$	0	1	3	4
$y:$	-12	0	6	12

23. a. Evaluate $\int_0^5 \frac{dx}{4x+5}$ by Simpson's 1/3 rule and hence find the value of $\log_e^5(n=10)$. 8 3 3 3

(OR)

b. Find $y'(x)$ given

x :	0	1	2	3	4
$y(x)$:	1	1	15	40	85

8 3 3 2

24. a. Using modified Euler's method solve, given that $y' = y - x^2 + 1, y(0) = 0.5$
Find $y(0.2)$.

8 3 4 2

(OR)

- b. Solve $y' = \frac{1}{2}(1 + x^2)y^2; y(0) = 1, y(0.1) = 1.06, y(0.2) = 1.12, y(0.3) = 1.21$
compute $y(0.4)$. Milne's predictor corrector formula.

8 3 4 3

25. a. Find the values of the function $u(x, t)$ satisfying $\frac{\partial u}{\partial t} = 4 \frac{\partial^2 u}{\partial x^2}$ and the
boundary conditions $u(0, t) = 0 = u(8, t)$ and $u(x, 0) = 4x - \frac{x^2}{2}$ at
 $x = i, i = 0, 1, 2, 3, 4$ and $t = \frac{1}{8}j; j = 0, 1, 2, 3, 4, 5$

8 4 5 3

(OR)

- b. Solve $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, 0 < x < 5, t > 0$ given that $u(x, 0) = 20, u(0, t) = 0,$
 $u(5, t) = 100$. Compute u for one time step with $h=1$ by crank Nicholson
method.

8 4 5 2

PART – C (1 × 15 = 15 Marks)

Answer ANY ONE Questions

Marks	BL	CO	PO
15	4	5	3

26. Solve $\nabla^2 u = 0$ over a square region of side 4 satisfying boundary
conditions.

- (i) $u(0, y) = 0$; $0 \leq y \leq 4$
- (ii) $u(4, y) = 12 + y$; $0 \leq y \leq 4$
- (iii) $u(x, 0) = 3x$; $0 \leq x \leq 4$
- (iv) $u(x, 4) = x^2$; $0 \leq x \leq 4$

By dividing the square into 16 sub squares of size 1 unit.

27. Find an equation of the cubic curve which passes through the points
(4, -43), (7, 83) (9, 327) and (12, 1058). Hence find $f(10)$ using Newton's
divided difference formula.

15 3 2 3

* * * * *

23. a. Find the values of $f'(4)$ and $f''(4)$ from the following data.

x:	0	1	2	3
y:	1	2	9	28

(OR)

b. Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using Trapezoidal rule with $h=0.2$. Hence find the value of π . Can you use Simpson's formula in this case?

24. a. Use Taylor's series method to find $y(1.1)$ correct to 4 decimal places given $y' = xy^{1/3}$ and $y(1) = 1$.

(OR)

b. Solve $y' = x^2 + y^2$; $y(0) = 1$ by modified Euler's method and find $y(0.2)$.

25. a. Solve $u_{xx} = 2u_t$, $u(0, t) = 0$, $u(4, t) = 0$ and $u(x, 0) = x(4 - x)$ choosing $h=k=1$ and using Bender Schmidt formula, find values upto $t=5$.

(OR)

b. Solve by Crank Nicholson method, $16u_t = u_{xx}$; $0 < x < 1$, $t > 0$ given that $u(x, 0) = 0$, $u(0, t) = 0$, $u(1, t) = 100t$.

PART - C (1 × 15 = 15 Marks)

Answer ANY ONE Question

Marks BL CO PO

26. Given $y'' + xy' + y = 0$; $y(0) = 1$, $y'(0) = 0$ find the value of $y(0.1)$ by R.K. method of fourth order.

27. If $f(0) = 0$, $f(1) = 0$, $f(2) = -12$, $f(4) = 0$, $f(5) = 600$, $f(7) = 7308$, find a polynomial that satisfies this data using Newton's divided difference formula, hence find $f(6)$.

Reg. No.

B.Tech. / M.Tech (Integrated) DEGREE EXAMINATION, MAY 2023 Third Semester

21MAB206T – NUMERICAL METHODS AND ANALYSIS
(For the candidates admitted from the academic year 2021 - 2022 & 2022 - 2023)

Note:

- Part - A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- Part - B and Part - C should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

PART - A (20 × 1 = 20 Marks)

Answer ALL Questions

Marks BL CO PO

- The negative root of the equation $e^x = 2 + x$ lies between
(A) -1.6 and -1.7 (B) -1.7 and -1.8
(C) -1.8 and -1.9 (D) -1.9 and -2.0
- The iteration formula for finding the reciprocal of N by N-R method is
(A) $x_{n+1} = x_n(2 - Nx_n)$ (B) $x_{n+1} = x_n(1 - Nx_n)$
(C) $x_{n+1} = x_n(1 + Nx_n)$ (D) $x_{n+1} = x_n(2 + Nx_n)$
- If the nth differences of a tabulated function are constants, then the function is a polynomial of degree
(A) 1 (B) 2
(C) n-1 (D) n
- $\Delta \log f(x) =$
(A) $\log f(x) - \log f(x+h)$ (B) $\log f(x+h) + \log f(x)$
(C) $\log f(x) + \log f(x-h)$ (D) $\log f(x+h) - \log f(x)$
- What is $\Delta^2 f(x)$, if $f(x) = x^2 - 3x + 1$ taking $h=1$
(A) $2x+1$ (B) $2x$
(C) 2 (D) 0
- The first divided difference of $f(x)$ for the arguments x_0, x_1 is defined as
(A) $\frac{f(x_1) - f(x_0)}{x_1 - x_0}$ (B) $\frac{f(x_1) - f(x_0)}{x_0 - x_1}$
(C) $\frac{f(x_1) - f(x_0)}{x_0 + x_1}$ (D) $\frac{f(x_0) - f(x_1)}{x_0 - x_1}$
- The relation between E and μ is
(A) $\mu = E^{1/2} + E^{-1/2}$ (B) $\mu = E^{-1/2} - E^{1/2}$
(C) $\mu = \frac{1}{2} [E^{1/2} + E^{-1/2}]$ (D) $\mu = \frac{1}{2} [E^{-1/2} - E^{1/2}]$

8. The central difference operator δ is given by
 (A) $\delta f(x) = f\left(x + \frac{h}{2}\right) + f\left(x - \frac{h}{2}\right)$ (B) $\delta f(x) = f(x) - f\left(x + \frac{h}{2}\right)$
 (C) $\delta f(x) = f(x) + f\left(x - \frac{h}{2}\right)$ (D) $\delta f(x) = f\left(x + \frac{h}{2}\right) - f\left(x - \frac{h}{2}\right)$
9. The error in the trapezoidal rule is of order
 (A) $\frac{h^2}{12}$ (B) $\frac{h^3}{6}$
 (C) $\frac{3h^2}{8}$ (D) h^2
10. Numerical differentiation is the process of computing the value of
 (A) Δy_0 (B) ∇y_0
 (C) $\frac{\Delta y}{\Delta x}$ (D) $\frac{dy}{dx}$
11. Simpson's 1/3 rule is also called
 (A) Parabolic rule (B) Hyperbolic rule
 (C) Elliptic rule (D) Trapezoidal rule
12. Simpson's 1/3 rule is $\int_{x_0}^{x_n} f(x) dx =$
 (A) $\frac{h}{3}[(y_0 + y_n) + 2(y_2 + y_4 + \dots) + 4(y_1 + y_3 + \dots)]$
 (B) $\frac{h}{2}[(y_0 + y_n) + 2(y_1 + y_2 + \dots)]$
 (C) $\frac{3h}{8}[(y_0 + y_n) + 2(y_2 + y_4 + \dots) + 4(y_1 + y_3 + \dots)]$
 (D) $\frac{h}{3}[(y_0 + y_n) + 2(y_1 + y_3 + \dots) + 4(y_2 + y_4 + \dots)]$
13. Which of the following method is better method?
 (A) Taylor's series method (B) Euler's method
 (C) R-K method (D) Modified Euler's method
14. How many prior values required to predict the next value in Adam's predictor corrector method?
 (A) 2 (B) 1
 (C) 3 (D) 4
15. Improved Euler's algorithm to find solution of a first order differential equation for $n=0,1,2,\dots$
 (A) $y_{n+1} = y_n + h\left[f\left(x_n + \frac{h}{2}, y_n\right) + \frac{h}{2}f(x_n, y_n)\right]$
 (B) $y_{n+1} = y_n + hf(x_n, y_n)$
 (C) $y_{n+1} = y_n + \frac{h}{2}[f(x_n, y_n) + f(x_n + h, y_n) + hf(x_n, y_n)]$
 (D) $y_{n+1} = y_n + \left[f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2}f(x_n, y_n)\right)\right]$

16. Adam's corrector formula is
 (A) $y_{n+1} = y_n + \frac{h}{24}[55y'_n - 58y'_{n-1} + 37y'_{n-2} - 9y'_{n-3}]$
 (B) $y_{n+1} = y_n + \frac{h}{24}[55y'_n + 59y'_{n-1} - 37y'_{n-2} + 9y'_{n-3}]$
 (C) $y_{n+1} = y_n + \frac{h}{24}[9y'_{n+1} + 19y'_n - 5y'_{n-1} + y'_{n-2}]$
 (D) $y_{n+1} = y_n + \frac{h}{24}[9y'_{n+1} - 19y'_n + 5y'_{n-1} + y'_{n-2}]$
17. If $B^2 - 4AC = 0$ then second order PDE is known as
 (A) Elliptic (B) Parabolic
 (C) Hyperbolic (D) Laplace equation
18. $\nabla^2 u = 0$ is called
 (A) Elliptic (B) Parabolic
 (C) Hyperbolic (D) Laplace equation
19. Bender Schmidt recurrence equation is valid only if
 (A) $k = \frac{h^2}{2}$ (B) $k = \frac{ah^2}{2}$
 (C) $k = \frac{2}{ah^2}$ (D) $k = \frac{2}{h^2}$
20. The error in the diagonal five point formula is _____ times the error in standard five point formula.
 (A) 1 (B) 2
 (C) 3 (D) 4

PART - B (5 × 8 = 40 Marks)

Answer ALL Questions

21. a. Find a negative root of the equation $x^3 - 2x + 5 = 0$ by the iteration method correct to 4 decimal places.

(OR)

- b. Find the positive root of the equation $3x - \cos x - 1 = 0$ by Newton's method correct to 4 decimal places.

22. a. Estimate the value of $f(22)$ from the following table:

x:	20	25	30	35	40	45
f(x):	354	332	291	260	231	204

(OR)

- b. Find $f(x)$ as a polynomial in x from the given data.

x:	3	7	9	10
f(x):	168	120	72	63

- b. Using Newton's forward interpolation, find the value of y at $x=43$.

x	40	50	60	70	80	90
y	184	204	226	250	276	304

30. a. Find the first, second and third derivatives of the function tabulated below at the point $x=1.5$

x	1.5	2	2.5	3	3.5	4
y	3.375	7.0	13.625	24.0	38.875	59.0

(OR)

- b. The velocity V of a particle at distances from a point on its path is given by the table.

S (feet)	0	10	20	30	40	50	60
V (feet/sec)	47	58	64	65	61	52	38

Estimate the time taken to travel 60 feet by using Simpson's $1/3^{\text{rd}}$ rule. Compare the result with Simpson's $3/8^{\text{th}}$ rule.

31. a. Given $\frac{dy}{dx} = x^2y - 1$ and $y(0) = 1$ find the value of $y(0.1)$ using Taylor series method.

(OR)

- b. Given $y'' + xy' + y = 0, y(0) = 1, y'(0) = 0$ find the value of $y(0.1)$ by using Runge-Kutta method of fourth order where $h=0.1$.

32. a. Solve the equation $\frac{\partial u}{\partial t} = \frac{1}{2} \frac{\partial^2 u}{\partial x^2}, 0 \leq x \leq 12, 0 \leq t \leq 12$ with boundary and initial conditions.

$$u(x, 0) = \frac{1}{4}x(15 - x), 0 \leq x \leq 12$$

$$u(0, t) = 0, u(12, t) = 9, 0 \leq t \leq 12$$

Using Schmidt relation.

(OR)

- b. Solve by Crank Nicholson's method

$$\frac{\partial u}{\partial t} = \frac{1}{16} \frac{\partial^2 u}{\partial x^2}, 0 < x < 1, t > 0, u(x, 0) = 0, u(0, t) = 0, u(1, t) = 100t$$

Compute u for one step with $h=1/4$.

* * * * *

Reg. No.

B.Tech. DEGREE EXAMINATION, MAY 2023

Fourth Semester

18MAB206T – NUMERICAL METHODS AND ANALYSIS

(For the candidates admitted from the academic year 2018-2019 to 2021-2022)

Note:

- (i) Part - A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) Part - B & Part - C should be answered in answer booklet.

Time: 3 hours

Max. Marks: 100

PART – A (20 × 1 = 20 Marks)

Answer ALL Questions

- | | Marks | BL | CO | PO |
|--|-------|----|----|----|
| 1. Newton – Raphson method is also known as the method of
(A) Tangents (B) Chords
(C) Diameter (D) Trapezoid | 1 | 1 | 1 | 1 |
| 2. The condition of convergence for iterative methods of solving a system of simultaneous linear equations is stated that the coefficient matrix should be
(A) Upper triangular (B) Diagonally dominant
(C) Conditionally convergent (D) Lower triangular | 1 | 1 | 1 | 1 |
| 3. The convergence in the Gauss-Seidel method is roughly _____ as faster as in Jacobi method.
(A) 3 times (B) 6 times
(C) 4 times (D) 2 times | 1 | 1 | 1 | 1 |
| 4. To find the inverse of matrix A, we use
(A) Gauss – elimination method (B) Gauss – Seidel method
(C) Gauss – Jacobi method (D) Newton – Raphson method | 1 | 1 | 1 | 1 |
| 5. The forward difference operator Δ is defined by
(A) $\Delta y_x = y_{x+1}$ (B) $\Delta y_x = y_{x+1} - y_{2x-1}$
(C) $\Delta y_x = y_{x+1} - y_x$ (D) $\Delta y_x = 2y_{x+1}$ | 1 | 1 | 2 | 1 |
| 6. The first divided difference of $f(x)$ for the arguments x_0 and x_1 is defined as
(A) $f(x_0, x_1) = \frac{x_1 - x_0}{f(x_1)}$ (B) $f(x_0, x_1) = x_1 - f(x_0) - x_2$
(C) $f(x_0, x_1) = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$ (D) $f(x_0, x_1) = \frac{f(x_1) - f'(x_0)}{x_1 - x_0}$ | 1 | 1 | 2 | 1 |
| 7. The differences $\Delta y_0, \Delta^2 y_0, \Delta^3 y_0 \dots$ are called the _____ differences.
(A) Leading (B) Diagonal
(C) Numerical (D) Differential | 1 | 1 | 2 | 1 |

8. Lagrange's interpolation formula is used for _____ intervals. 1 1 2 1
 (A) Equal (B) Unequal
 (C) Both equal and unequal (D) Neither equal nor unequal
9. Close approximation in Simpson's one-third rule is obtained when h is 1 1 3 1
 (A) Small (B) Large
 (C) Infinity (D) Greater
10. Trapezoidal rule approximates the integral by sum of n 1 1 3 1
 (A) Rhombus (B) Circle
 (C) Triangle (D) Trapezoids
11. Simpson's one-third rule is also called 1 1 3 1
 (A) Hyperbolic rule (B) Elliptic rule
 (C) Parabolic rule (D) Trapezoid rule
12. Simpson's three-eight rule can be applied only when n is 1 1 3 1
 (A) Odd (B) Even
 (C) Prime (D) Multiple of 3
13. Taylor's series method is 1 1 4 1
 (A) Single step method (B) Multi step method
 (C) Iterative method (D) Error method
14. The improved Euler method is based on the average of 1 1 4 1
 (A) Points (B) Slopes
 (C) Curves (D) Errors
15. Consider the function $y = f(x)$ such that $f(1) = 5$ and $f(3) = 11$. The 1 2 3 1
 estimated value of $\int_1^3 f(x) dx$ with $h=2$ using Trapezoidal rule is
 (A) 4 (B) 8
 (C) 12 (D) 16
16. Which of the following is a method for finding a numerical integration? 1 1 3 2
 (A) Euler's method (B) Tangent method
 (C) Simpson's one-third rule (D) Jacobi method
17. If $B^2 - 4AC < 0$, then the second order PDE is known as 1 1 5 1
 (A) Elliptic (B) Parabolic
 (C) Hyperbolic (D) Wave equation
18. The equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$ is known as 1 1 5 1
 (A) Elliptic (B) Laplace equation
 (C) Poisson equation (D) Wave equation
19. The error in the diagonal formula is _____ times the error in the standard 1 1 5 1
 formula.
 (A) One (B) Two
 (C) Three (D) Four

20. The nature of the partial differential equation $f_{xy} - f_x = 0$ is 1 1 5 1
 (A) Elliptic (B) Parabolic
 (C) Hyperbolic (D) Laplace equation

PART - B (5 × 4 = 20 Marks)

Answer ANY FIVE Questions

21. Evaluate $\sqrt{12}$ to four decimal places by Newton's-Raphson's method. 4 6 1 2

22. Solve the system of equations by Gauss elimination method: 4 3 1 2
 $x + 2y + z = 3, 2x + 3y + 3z = 10, 3x - y + 2z = 13$.

23. Express $x^3 + x^2 + x + 1$ in factorial polynomial and get their successive 4 2 2 2
 forward differences taking $h=1$.

24. Find the missing term in the following table. 4 4 2 2

x	1	2	3	4	5	6	7
y	2	4	8	-	32	64	128

25. Evaluate the integral $I = \int_4^{5.2} \log_e^x dx$ using Trapezoidal rule. 4 6 3 2

26. Compute y at $x=0.25$ by Modified-Euler method given $y' = 2xy, y(0) = 1$. 4 3 4 2

27. Classify the following partial differential equation 4 4 5 2
 (i) $xf_{xx} + yf_{yy} = 0, x > 0, y > 0$
 (ii) $f_{xx} - 2f_{xy} = 0, x > 0, y > 0$

PART - C (5 × 12 = 60 Marks)

Answer ALL Questions

28. a. Solve the following system of equations by using Gauss-Jacobi method 12 3 1 2
 correct to 3 decimal places

$$8x - 3y + 2z = 20$$

$$4x + 11y - z = 33$$

$$6x + 3y + 12z = 35$$

(OR)

- b. Solve by Gauss Seidel method the following system 12 3 1 2

$$28x + 4y - z = 32$$

$$x + 3y + 10z = 24$$

$$2x + 17y + 4z = 35$$

29. a. Using Lagrange's formula, fit a polynomial to the data. Also find y at $x=2$. 12 3 2 2

x	0	1	3	4
y	-12	0	6	12

(OR)

- b. Analyze the following system of equations and solve it by Gauss-Seidel method.
 $4x + 2y + z = 14$
 $x + 5y - z = 10$
 $x + y + 8z = 20$

27. a. The population of a town in the census is given below. Applying the interpolation technique, estimate the population in the year 1895 and 1925.

Year (x):	1891	1901	1911	1921	1931
Population (y): (in 1000's)	46	66	81	93	101

(OR)

- b. Determine the polynomial $y(x)$ for the following data using Lagrange's interpolation formula.

x	0	1	4	5
y(x)	4	3	24	39

28. a. In a machine, a slider moves along a fixed straight rod. Its distance x cms. along the rod is given below for various values of time t sec.

t:	0.1	0.2	0.3	0.4	0.5
x:	31.62	32.87	33.64	33.95	33.81

Analyse the technique of numerical differentiation and determine the velocity of the slider when $t=0.1$ and $t=0.5$.

(OR)

- b. The velocity of the train at various times are given by the following table,

Minutes:	0	2	4	6	8	10	12	14	16	18	20
Miles/hr:	0	10	18	25	29	32	20	11	5	2	0

Apply numerical integration to calculate the total distance covered in 20 minutes.

29. a.i. Given $y' = xy + y^2$, $y(0) = 1$, use Taylor series method to estimate the value of y at $x=0.1$, taking $h=0.1$.

- ii. Use Modified Euler method to calculate $y(0.2)$, given $y' = y - x^2$, $y(0) = 1$ by taking $h=0.2$.

(OR)

- b. Apply Milne's predictor-corrector method to estimate the value of $y(0.4)$, given $y' = 2e^x - y$, $y(0) = 2$, $y(0.1) = 2.010$, $y(0.2) = 2.040$, $y(0.3) = 2.090$, by analyzing the predictor and corrector formula.

30. a. Solve $\nabla^2 u = -10(x^2 + y^2 + 10)$ over the square mesh with sides $x=0$, $y=0$, $x=3$, $y=3$ with $u=0$ on the boundary and mesh length 1 unit, by using suitable numerical technique.

(OR)

- b. Analyse the one half period of vibration and evaluate the pivotal values of the following equations taking $h=1$ for one half period of vibration, given

$$u(0,t)=0, u(5,t)=0, u_t(x,0)=0 \text{ and } u(x,0)=\begin{cases} 2x & , 0 \leq x \leq 2.5 \\ 10-2x & , 2.5 \leq x \leq 5 \end{cases}$$

* * * * *

Reg. No.

B.Tech. DEGREE EXAMINATION, MAY 2022
 Fourth Semester

18MAB202T – NUMERICAL METHODS FOR ENGINEERS

(For the candidates admitted from the academic year 2018-2019 to 2019-2020)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
 (ii) **Part - B** should be answered in answer booklet.

Time: 2½ Hours

Max. Marks: 75

PART – A (25 × 1 = 25 Marks)

Answer ALL Questions

- | | Marks | BL | CO | PO |
|--|-------|----|----|----|
| 1. When we fit a parabola by the method of least squares, which of the following is the formula to calculate the error?
(A) $\sum y^2 + a \sum x^2 y + b \sum xy + c \sum y$ (B) $\sum y^2 - a \sum x^2 y - b \sum xy - c \sum y$
(C) $\sum y^2 - a \sum xy^2 - b \sum xy - c \sum x$ (D) $\sum y^2 - a \sum x^2 y + b \sum xy - c \sum x$ | 1 | 1 | 1 | 1 |
| 2. In solving simultaneous equations by Gauss Jordan method, the co-efficient matrix is reduced to
(A) Null matrix (B) Diagonal matrix
(C) Upper triangular matrix (D) Lower triangular matrix | 1 | 1 | 1 | 1 |
| 3. Which of the following is one of the normal equation to fit a straight line $y=ax+b$?
(A) $\sum y = a \sum x + nb$ (B) $\sum xy = a \sum x - nb$
(C) $\sum y^2 = a \sum y + nb$ (D) $\sum y = a \sum xy + b \sum x$ | 1 | 1 | 1 | 1 |
| 4. Find the range, in which the negative root lies while solving $x^3 - 2x + 5 = 0$
(A) 0 and 1 (B) -1 and -2
(C) -2 and -3 (D) 1 and 2 | 1 | 2 | 1 | 2 |
| 5. Write the first iteration values of x and y, by Gauss-Seidel method for the system of equations $4x + y = 2$; $2x + 3y = 5$
(A) $x=4, y=3$ (B) $x=1, y=2$
(C) $x=0.5, y=1.67$ (D) $x=0.5, y=1.33$ | | | | |
| 6. The operator E is equivalent to
(A) $(1 + \Delta)^{-1}$ (B) $(1 - \Delta)^{-1}$
(C) $(1 + \nabla)^{-1}$ (D) $(1 - \nabla)^{-1}$ | 1 | 1 | 2 | 1 |
| 7. The first divided difference of $f(x)$ for the arguments x_0, x_1 is defined as
(A) $\frac{f(x_1) - f(x_0)}{x_1 - x_0}$ (B) $\frac{f(x_1) - f(x_0)}{x_0 - x_1}$
(C) $\frac{f(x_1) + f(x_0)}{x_1 + x_0}$ (D) $\frac{f(x_0) + f(x_1)}{x_1 - x_0}$ | 1 | 1 | 2 | 1 |

8. If $3x^{(3)} + 7x^{(2)} + 8x^{(1)} - 6$ is the factorial, polynomial, then find the second forward difference, by taking $h=1$.
 (A) $9x^{(2)} - 14x^{(1)} + 8$ (B) $18x^{(1)} - 14$
 (C) $18x^{(1)} + 14$ (D) $9x^{(2)} + 14x^{(1)} + 8$

9. The polynomial that suits the given data $f(0)=4, f(1)=3$ and $f(4)=24$, by suitable interpolation technique is
 (A) $x^2 - 2x + 4$ (B) $-3x^2 + 2x + 4$
 (C) $2x^2 - 3x + 4$ (D) $-2x^2 + x + 4$

10. The third differences of $f(x)$ for the following data is

x	0	1	2	3	4
f(x)	1	3	7	13	21

 (A) 0,0 (B) 0,1
 (C) 1,1 (D) 1,0

11. Which of the following is the forward difference formula to get the first derivative at $x=x_0$?
 (A) $\frac{1}{h} \left(\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \dots \right)$ (B) $\frac{1}{h^2} \left(\Delta^2 y_0 - \frac{1}{2} \Delta^3 y_0 + \dots \right)$
 (C) $\frac{1}{h} \left(\Delta y_0 + \frac{1}{2} \Delta^2 y_0 + \dots \right)$ (D) $\frac{1}{h^2} \left(\Delta^2 y_0 + \frac{1}{2} \Delta^3 y_0 + \dots \right)$

12. The accuracy in the trapezoidal rule can be improved by
 (A) Increasing the number of intervals (B) Increasing the value of h
 (C) Decreasing the number of intervals (D) Without changing the value of h

13. Simpson's three-eight rule can be applied only when the number of intervals is
 (A) Odd (B) even
 (C) Multiple of 3 (D) Multiple of 2

14. Using the following table, find the value of $f'(1)$

x	1	2	3
f(x)	4	26	48

 (A) -7 (B) 22
 (C) 11 (D) 7

15. Find the value of $\int_{1/2}^1 \frac{1}{x} dx$ by trapezoidal rule by dividing the range into 4 equal parts.
 (A) 0.527 (B) 0.321
 (C) 0.490 (D) 0.697

16. Taylor's series method is known as
 (A) Single step method (B) Multi step method
 (C) Iterative method (D) Trial and error method

17. The improved Euler method is based on the averages of
 (A) Points (B) Slopes
 (C) Curves (D) Both points and slopes

18. How many prior values are required to predict the next value in predictor-corrector methods?
 (A) 1 (B) 2
 (C) 3 (D) 4

19. Given $y' = x + y, y(0) = 1$, find the value of $y(0.1)$ by Euler's method.
 (A) 1 (B) 0.2
 (C) 1.1 (D) 0.1

20. Using Runge-Kutta method of fourth order, if $k_1=0.2, k_2=0.205, k_3=0.2052, k_4=0.2105$ then the value of Δy will be
 (A) 0.2052 (B) 0.3053
 (C) 0.1038 (D) 1.2052

21. If $B^2 - 4AC = 0$, then the second order linear PDE is classified as
 (A) Cyclic (B) Hyperbolic
 (C) Parabolic (D) Elliptic

22. The PDE $xf_{xx} + yf_{yy} = 0, x < 0, y > 0$ is classified as
 (A) Parabolic (B) Cyclic
 (C) Elliptic (D) Hyperbolic

23. The term $u_{ij} = \frac{1}{4} [u_{i-1,j-1} + u_{i-1,j+1} + u_{i+1,j-1} + u_{i+1,j+1}]$ is known as
 (A) Diagonal five point formula (B) Standard five point formula
 (C) Explicit formula (D) Implicit formula

24. Bender-Schmidt recurrence equation is valid only if
 (A) $k = \frac{ah^2}{2}$ (B) $k = \frac{h^2}{2}$
 (C) $k = \frac{2}{h^2}$ (D) $k = \frac{2}{ah^2}$

25. Crank-Nicholson recurrence scheme is used to solve
 (A) One dimensional Wave equation (B) One dimensional Heat equation
 (C) Laplace equation (D) Poisson equation

PART - B (5 × 10 = 50 Marks)

Answer ALL Questions

26. a.i. Using the method of least squares, fit a straight line to the following data.

x	3	4	5	6
y	6	9	10	11

- ii. Determine the root of $f(x) = \cos x - xe^x$, using Newton-Raphson method, correct to 4 decimal places.

(OR)

- b. Apply Lagrange's interpolation formula to fit a polynomial to the data given below. Also, find y at $x=2$.

x	1	3	4
y	0	6	12

30. a. In a machine a slider moves along a fixed straight rod. Its distance x cms along the rod is given below for various values of time t secs. Estimate the velocity and acceleration of the slider when $t=0.3$

t (sec):	0.0	0.1	0.2	0.3	0.4	0.5
x (cm):	29.91	31.63	32.87	33.64	33.95	33.81

(OR)

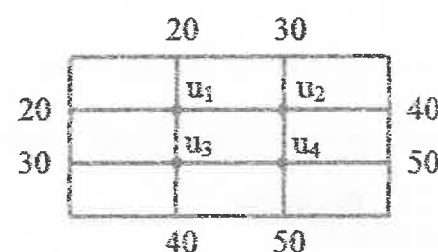
- b. Apply trapezoidal and Simpson's rule to evaluate $\int_0^{\pi} \sin x dx$, by dividing the range into ten equal parts.

31. a. Given that $y' = x + x^2 y$, $y(0) = 1$. Compute $y(0.1)$ and $y(0.2)$ by applying Euler's method and R-K method of fourth order respectively.

(OR)

- b. Applying Milne's predictor-corrector method, compute $y(4.4)$, given $5xy' + y^2 - 2 = 0$, $y(4) = 1$, $y(4.1) = 1.0049$, $y(4.2) = 1.0097$, $y(4.3) = 1.0143$.

32. a. Solve $\nabla^2 u = 0$ for square region with the given boundary conditions using numerical techniques.



(OR)

- b.i. Solve $u_{xx} = 16u_t$, $0 < x < 1$, $t > 0$ given $u(x, 0) = 0$, $u(0, t) = 0$, $u(1, t) = 100t$. Compute u for one step in t direction taking $h=0.25$ using Crank-Nicholson scheme.

- ii. Solve numerically, $4u_{xx} = u_{tt}$ with the given conditions $u(0, t) = 0$, $u(4, t) = 0$, $u_t(x, 0) = 0$, $u(x, 0) = x(4 - x)$, taking $h=1$ upto $t=2$ secs.

Reg. No.

B.Tech. DEGREE EXAMINATION, MAY 2023

Fourth Semester

18MAB202T – NUMERICAL METHODS FOR ENGINEERS

(For the candidates admitted from the academic year 2018-2019 to 2021-2022)

Note:

- (i) Part - A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) Part - B & Part - C should be answered in answer booklet.

Time: 3 hours

Max. Marks: 100

PART - A (20 × 1 = 20 Marks)

Answer ALL Questions

- In fitting a straight line by the method of least squares, the error committed is given by
(A) $E = \sum y^2 - a \sum xy - b \sum y$ (B) $E = \sum y + a \sum xy - b \sum y^2$
(C) $E = \sum y^2 - a \sum x^2 y - b \sum xy$ (D) $E = \sum y^2 - a \sum xy + b \sum y$
- In solving simultaneous linear equation by Gauss-Jordan method, the co-efficient matrix is reduced to
(A) Lower triangular matrix (B) Diagonal matrix
(C) Upper triangular matrix (D) Null matrix
- Write the condition for convergence in Newton-Raphson method.
(A) $|f(x)f''(x)| > |f'(x)|^2$ (B) $|f(x)f'(x)| > |f''(x)|^2$
(C) $|f(x)f''(x)| < |f'(x)|^2$ (D) $|f(x)f'(x)|^2 > |f''(x)|$
- Power method is used to determine the
(A) Smallest eigen value (B) Positive eigen values
(C) Negative eigen value (D) Numerically largest eigen value
- The operator E is equivalent to
(A) $(1 + \Delta)^{-1}$ (B) $(1 - \Delta)^{-1}$
(C) $(1 + \nabla)^{-2}$ (D) $(1 - \nabla)^{-1}$
- The missing term in the following table using finite difference technique is

x	0	1	2	3	4
$f(x)$	1	3	9	-	81

(A) 27 (B) 31
(C) 30 (D) 29
- Find the factorial polynomial for the equation $3x^3 - 2x^2 + 7x - 6$, by taking $h=1$.
(A) $3x^{(3)} + x^{(2)} + 8x^{(1)} + 6$ (B) $3x^{(3)} + 7x^{(2)} + 8x^{(1)} - 6$
(C) $x^{(3)} + 7x^{(2)} - 8x^{(1)} - 6$ (D) $2x^{(3)} + 7x^{(2)} + 8x^{(1)} + 6$
- The first divided difference of $y(x)$ for the arguments x_0, x_1 is given by
(A) $\frac{y_1 - y_0}{x_1 - x_0}$ (B) $\frac{y_1 + y_0}{x_1 + x_0}$
(C) $\frac{y_2 - y_0}{x_2 - x_0}$ (D) $\frac{y_2 + y_1}{x_2 + x_1}$

9. Newton's forward difference formula to get the first order derivative at $x=x_0$ is
- (A) $\frac{dy}{dx} = \frac{1}{h} \left[\nabla y_n - \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n - \dots \right]$
- (B) $\frac{dy}{dx} = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \dots \right]$
- (C) $\frac{dy}{dx} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \dots \right]$
- (D) $\frac{dy}{dx} = \frac{1}{h} \left[\Delta y_0 + \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 + \dots \right]$
10. The error in the trapezoidal rule is
- (A) $|E| < \frac{(b-a)^2}{12} h^4$
- (B) $|E| < \frac{(b-a)}{24} h^4$
- (C) $|E| < \frac{(b-a)^2}{24} h^2$
- (D) $|E| < \frac{(b-a)}{12} h^2$
11. The error in Simpson's one-third rule is of order
- (A) h^6
- (B) h^3
- (C) h^4
- (D) h^2
12. Simpson's three-eighth rule can be applied only when the number of intervals is
- (A) Multiple of 3
- (B) Even
- (C) Odd
- (D) Any number
13. Which of the following is a multi-step method?
- (A) Euler method
- (B) Adam's predictor corrector method
- (C) Taylor's method
- (D) Runge-kutta method
14. The modified Euler method is based on the averages of
- (A) Points
- (B) Slopes
- (C) Centres
- (D) Tangents
15. How many prior values are required to predict the next value in Milne's method?
- (A) 1
- (B) 2
- (C) 4
- (D) 3
16. If $y' = x + y$, $y(0) = 1$, $h = 0.2$, then the value of $y(0.2)$ using Euler's method is
- (A) 1
- (B) 2
- (C) 1.5
- (D) 1.2
17. The nature of the partial differential equation $f_{xx} - 2f_{xy} = 0$ is
- (A) Hyperbolic
- (B) Elliptic
- (C) Parabolic
- (D) Cyclic
18. The partial differential equation $\frac{\partial^2 u}{\partial x^2} = \frac{1}{\alpha^2} \frac{\partial u}{\partial t}$ is classified as
- (A) Elliptic
- (B) Parabolic
- (C) Hyperbolic
- (D) Cyclic
19. Bender-Schmidt recurrence equation is valid only if
- (A) $k = \frac{ah}{2}$
- (B) $k = \frac{a^2 h}{4}$
- (C) $k = \frac{ah^2}{3}$
- (D) $k = \frac{ah^2}{2}$

20. The partial differential equation $\nabla^2 u = f(x, y)$ is called as
- (A) Poisson equation
- (B) Laplace equation
- (C) Parabolic equation
- (D) Hyperbolic equation

PART - B (5 × 4 = 20 Marks)

Answer ANY FIVE Questions

21. Using Newton-Raphson method derive the iterative formula to find $\sqrt{N}$, where N is a positive integer.
22. Find the value of x and y for the following system of equation using Gauss-Elimination method.
- $$11x + 3y = 17$$
- $$2x + 7y = 16$$
23. Form the divided difference table for the following data.
- | | | | | | | |
|---|----|-----|-----|-----|------|------|
| x | 4 | 5 | 7 | 10 | 11 | 13 |
| y | 48 | 100 | 294 | 900 | 1210 | 2028 |
24. The velocity v of a particle moving in a straight line covers a distance x in time t. They are related as follows.
- | | | | | | |
|---|----|----|----|----|----|
| x | 0 | 10 | 20 | 30 | 40 |
| v | 45 | 60 | 65 | 54 | 42 |
- Estimate the time taken to traverse the distance of 40 units by Simpson's 1/3 rule.
25. Solve $\frac{dy}{dx} = x + y$, given $y(1)=0$ and get $y(1.1)$ by Taylor series method.
26. Using improved Euler method compute y at $x=0.1$ given that $\frac{dy}{dx} = y - \frac{2x}{y}$, $y(0)=1$.
27. Solve $\frac{\partial^2 u}{\partial x^2} - 2 \frac{\partial u}{\partial t} = 0$ given $u(0, t) = 0$, $u(4, t) = 0$, $u(x, 0) = x(4 - x)$. Assume $h=1$. Estimate the values of u upto $t=5$, using Bender-Schmidt's method.

PART - C (5 × 12 = 60 Marks)

Answer ALL Questions

28. a. Fit a straight line and parabola by the method of least square for the following data and analyze which curve is more appropriate.
- | | | | | | |
|---|---|---|----|----|----|
| x | 0 | 1 | 2 | 3 | 4 |
| y | 1 | 5 | 10 | 22 | 38 |
- (OR)
- b. Solve the following system of equation by Gauss-Seidel method.
- $$4x + 2y + z = 14$$
- $$x + 5y - z = 10$$
- $$x + y + 8z = 20$$
29. a. The following table gives the marks got by 100 students in mathematics examination. Estimate how many students got more than 55 marks.
- | | | | | | |
|------------------|-------|-------|-------|-------|-------|
| Marks | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 |
| No. of students: | 25 | 35 | 22 | 11 | 7 |

(OR)

B. Tech. (PT) DEGREE EXAMINATION, JULY 2023**Second Semester****19PMAB02T – NUMERICAL METHODS FOR ENGINEERS***(For the candidates admitted during the academic year 2019 – 2020 onwards)*

Time: Three Hours

Max. Marks: 100

Answer **ALL** Questions**PART – A (10 × 2 = 20 Marks)**

- | | Marks | BL | CO | PO |
|---|-------|----|----|----|
| 1. Write the normal equations to fit a line $y = ax + b$ under the least square method. | 2 | 1 | 1 | 2 |
| 2. State diagonally dominant condition for Gauss Seidel method. | 2 | 1 | 1 | 2 |
| 3. State Newton's forward interpolation formula. | 2 | 1 | 2 | 2 |
| 4. What is the Lagrange's formula to find 'y' if three sets of values (x_0, y_0) , (x_1, y_1) and (x_2, y_2) are given? | 2 | 1 | 2 | 2 |
| 5. Write the Newton's formula for $\frac{dy}{dx}$ at $x = x_0$ and $x = x_n$. | 2 | 1 | 3 | 2 |
| 6. Write the order of error term in Trapezoidal rule. | 2 | 1 | 3 | 2 |
| 7. Using Euler's method, find $y(0.1)$ given that $\frac{dy}{dx} = -y$, $y(0) = 1$. | 2 | 1 | 4 | 1 |
| 8. Write Adams Bosh Fourth Predictor Corrector formula. | 2 | 1 | 4 | 2 |
| 9. What is the purpose of Liebmann's iterative formula? | 2 | 1 | 5 | 2 |
| 10. Classify the equation $\frac{\partial^2 u}{\partial x^2} + 2\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$. | 2 | 2 | 5 | 1 |

PART – B (5 × 16 = 80 Marks)

- 11.a. Fit a straight line and parabola to the following data.

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

(OR)

- | | Marks | BL | CO | PO |
|---|-------|----|----|----|
| b. Solve the following system by Gauss-Jacobi method.
$10x - 5y - 2z = 3$, $4x - 10y + 3z = -3$ and $x + 6y + 10z = -3$ | 16 | 3 | 1 | 2 |
| 12.a. The population of a town is as follows: | 16 | 4 | 2 | 2 |
- | Year (x) | 1941 | 1951 | 1961 | 1971 | 1981 | 1991 |
|-------------------------|------|------|------|------|------|------|
| Population in lakhs (y) | 20 | 24 | 29 | 36 | 46 | 51 |
- Estimate the population increase during the period 1946 to 1976.

(OR)

- b. Using Lagrange's interpolation formula, calculate
- $y(10)$
- from the following table.

x	5	6	9	11
y	12	13	14	16

- 13.a. The population of a certain town is given below. Find the rate of growth of the population in 1931 and 1961. 16 3 3 2

Year (x)	1931	1941	1951	1961	1971
Population in thousands (y)	40.62	60.80	79.95	103.56	132.65

(OR)

- b. By dividing the range into ten equal parts, evaluate $\int_0^{\pi} \sin x dx$ by Trapezoidal and Simpson's rule. Verify your answer with actual integration. 16 4 3 2

- 14.a. Apply the fourth order Runge-Kutta method to find $y(0.2)$ given that $\frac{dy}{dx} = x + y, y(0) = 1$. 16 3 4 2

(OR)

- b. Given $\frac{dy}{dx} = \frac{1}{2}(1+x^2)y^2$ and $y(0)=1, y(0.1)=1.06, y(0.2)=1.12, y(0.3)=1.21$, evaluate $y(0.4)$ by Milne's Predictor-Corrector method. 16 3 4 2

- 15.a. Solve $u_{xx} + u_{yy} = 0$ over a square mesh of side 4 units; satisfying the following boundary conditions. 16 3 5 2

- (i) $u(0, y) = 0$ for $0 \leq y \leq 4$
- (ii) $u(4, y) = 12 + y$, for $0 \leq y \leq 4$
- (iii) $u(x, 0) = 3x$ for $0 \leq x \leq 4$
- (iv) $u(x, 4) = x^2$ for $0 \leq x \leq 4$

(OR)

- b. Solve $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$ given $u(0, t) = 0, u(4, t) = 0, u(x, 0) = x(4 - x)$ assuming $h=k=1$. Find the value of 'u' upto $t=5$ using Bender-Schmidt formula. 16 3 5 2

* * * *

[illegible]

B. Tech. (PT) DEGREE EXAMINATION, JULY 2023
Second Semester

19PMAB04T – PROBABILITY AND STOCHASTIC PROCESSES

(For the candidates admitted during the academic year 2019 – 2020 onwards)

Time: Three Hours

Max. Marks: 100

Answer ALL Questions

PART – A (10 × 2 = 20 Marks)

Marks	BL	CO	PO
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- | | | | | |
|--|---|---|---|---|
| 1. Define discrete random variable with example. | 2 | 1 | 1 | 1 |
| 2. In a binomial distribution the mean is 4 and variance is 3. Find $P(X=0)$. | 2 | 2 | 1 | 1 |
| 3. The joint p.d.f of two random variable X and Y is given by | 2 | 2 | 2 | 2 |

$$f_{xy}(x,y)=\frac{1}{8}x(x-y), 0 < x < 2, -x < y < x$$

Find the marginal density function of X.

- | | | | | |
|---|---|---|---|---|
| 4. Define co-variance. | 2 | 1 | 2 | 1 |
| 5. Define Markov's inequality. | 2 | 1 | 3 | 1 |
| 6. Define Lindberg-Levi form. | 2 | 1 | 3 | 1 |
| 7. Define WSS process. | 2 | 1 | 4 | 1 |
| 8. Define Markov process. | 2 | 1 | 4 | 1 |
| 9. Given $S_{XX}(\omega) = \frac{1}{4 + \omega^2}$, find the average power of the process. | 2 | 2 | 5 | 1 |
| 10. State Wiener-Khintchine theorem. | 2 | 1 | 5 | 1 |

PART – B (5 × 16 = 80 Marks)

Marks	BL	CO	PO
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- 11.a. A random variable X has the following probability function.

x	0	1	2	3	4	5	6	7
P(x)	0	k	2k	2k	3k	k ²	2k ²	7k ² +k

- (i) Find k
- (ii) Evaluate $P(X < 6)$, $P(X \geq 6)$
- (iii) Determine the CDF

(OR)

- b. Find the area under the standard normal curve which lie
- (i) To the right of $Z = 2.70$
 - (ii) To the left of $Z = 1.73$
 - (iii) Between $Z = -1.45$ and $Z = 1.45$
 - (iv) Between $Z = -0.90$ and $Z = 0.58$

- 12.a. If the joint pdf of (x,y) is given by $f(x,y) = k(2x+3y)$, $x=0,1,2; y=1,2,3$. Find the all marginal probability distribution. Also find the distribution of (X+Y).

(OR)

b. Obtain the rank correlation coefficient for the following data.

16 3 2 2

X	38	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

13.a. A fair dice is tossed 720 times. Use Chebyshev's inequality to find a lower bound for the probability of getting 100 to 140 sixes.

16 3 3 2

(OR)

b. If $X_1, X_2, \dots, X_n$ are Poisson variates with parameter $\lambda=2$, using central limit theorem, find $P(120 \leq S_n \leq 160)$ where $S_n = X_1 + X_2 + \dots + X_n$ and $n=75$.

16 3 3 2

14.a. Give a random variable Y with characteristic function

16 2 4 2

$\phi(\omega) = E(e^{i\omega Y}) = E(\cos \omega Y + i \sin \omega Y)$ and random process defined by $X(t) = \cos(\lambda t + Y)$. Show that X(t) is stationary in the wide sense if $\phi(1) = \phi(2) = 0$.

(OR)

b. If $X(t) = 5 \cos(10t + \theta)$ and $Y(t) = 20 \sin(10t + \theta)$, where θ is a random variable uniformly distributed in $(0, 2\pi)$. show that the process X(t) and Y(t) are jointly wide sense stationary.

16 2 4 2

15.a. Given the power spectral density of a continuous process as

16 2 5 2

$S_{XX}(\omega) = \frac{\omega^2 + 9}{\omega^4 + 5\omega^2 + 4}$. Find the mean square value of the process.

(OR)

b. The auto correlation function of the random binary transmission is given by

16 2 5 2

$$R_{XX}(\tau) = \begin{cases} 1 - \frac{|\tau|}{T}, & \text{for } |\tau| \leq T \\ 0, & \text{for } |\tau| > T \end{cases}$$

Find the power spectrum.

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B.Tech. DEGREE EXAMINATION, DECEMBER 2022
Fourth Semester

18MAB202T – NUMERICAL METHODS FOR ENGINEERS
(For the candidates admitted from the academic year 2018-2019 to 2021-2022)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part - B** should be answered in answer booklet.

Time: 2½ Hours

Max. Marks: 75

PART – B (5 × 10 = 50 Marks)

Answer **ALL** Questions

Marks BL CO PO

26. a. Fit a parabola by the method of least square to the data given below. Also find y at x=1.5. 10 3 1 1

x	1	2	3	4	5
y	5	12	26	60	97

(OR)

- b. Solve the system of equations by Gauss- Seidel method to four decimal places 10 3 1 1
 $28x+4y - z = 32$, $x +3y + 10z = 24$, $2x + 17y + 4z = 35$.

27. a. The population of a town is as follows. 10 3 2 2

Year : x	1941	1951	1961	1971	1981	1991
Population in lakhs : y	20	24	29	36	46	51

Estimate the population increase during the period 1946 to 1976.

(OR)

- b. Using Lagrange's interpolation formula find y(1) from the following table. 10 3 2 1

x	-1	0	2	3
y	-8	3	1	12

28. a. Find $y'(51)$ and $y''(51)$ from the data given below 10 3 3 1

x	50	60	70	80	90
y	19.96	36.65	58.81	77.21	94.61

(OR)

- b. The velocity V of a particle at distance S from a point on its path is given by the table below. 10 3 3 1

S in metres	0	10	20	30	40	50	60
V m/sec	47	58	64	65	61	52	38

Estimate the time taken to travel 60 metres by using Simpson's one third and three eighth rules.

29. a. Solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ given $y(0.2) = 1.1960$ at $x = 0.4$ using Runge-Kutta fourth order method. 10 3 4 1

(OR)

b. Solve $\frac{dy}{dx} = y(x + y)$, $y(0) = 1$, $y(0.1) = 1.1169$, $y(0.2) = 1.2774$ and $y(0.3) = 1.5041$ at $x = 0.4$ by using Adam's method. 10 3 4 1

30. a. Solve $u_{xx} + u_{yy} = 0$ over the square mesh of sides 4 units, satisfying the following boundary conditions: (Divide the square mesh into 16 sub squares) 10 3 5 1

- (i) $u(0, y) = 0$ for $0 \leq y \leq 4$
- (ii) $u(4, y) = 12 + y$ for $0 \leq y \leq 4$
- (iii) $u(x, 0) = 3x$ for $0 \leq x \leq 4$
- (iv) $u(x, 4) = x^2$ for $0 \leq x \leq 4$

(OR)

b. Solve $25u_{xx} = u_{tt}$ for u at the pivotal points, given 10 3 5 1

$u(0, t) = u(5, t) = 0$, $u_t(x, 0) = 0$ and $u(x, 0) = \begin{cases} 2x & \text{for } 0 \leq x \leq 2.5 \\ 10 - 2x & \text{for } 2.5 \leq x \leq 5. \end{cases}$

for one half period of vibration.

* * * * *

29. a. From the following data find θ using Newton's interpolation formula at $x = 43$ and $x = 84$.

x	40	50	60	70	80	90
θ	184	204	226	250	276	304

Also express θ in terms of x .

(OR)

- b. Find the age corresponding to the annuity value $y = 13.6$ given the table, using Lagrange's interpolation formula.

Age (x)	30	35	40	45	50
Annuity value (y)	15.9	14.9	14.1	13.3	12.5

30. a. The table below gives the results of an observation; θ is the observed temperature in degrees centigrade of a vessel of cooling water; t is the time in minutes from the beginning of observation.

t	1	3	5	7	9
θ	85.3	74.5	67.0	60.5	54.3

Find the approximate rate of cooling at $t=3$ and $t = 3.5$.

(OR)

- b. Evaluate the integral $I = \int_4^{5.2} \log_e x \, dx$ using Simpson's rule and Trapezoidal rule.

31. a. Using Taylor's series method, find y at $x = 0.1$ (0.1) 0.4 given $\frac{dy}{dx} = x^2 - y$, $y(0) = 1$, correct to four decimal places.

(OR)

- b. Compute $y(0.3)$ given $\frac{dy}{dx} + y + xy^2 = 0$, $y(0) = 1$, by taking $h = 0.1$, using Runge Kutta method of fourth order.

32. a. Solve $u_{xx} + u_{yy} = 0$ over the square mesh of side 4 units; satisfying the following boundary conditions

- $u(0, y) = 0$ for $0 \leq y \leq 4$
- $u(4, y) = 12 + y$ for $0 \leq y \leq 4$
- $u(x, 0) = 3x$, for $0 \leq x \leq 4$
- $u(x, 4) = x^2$ for $0 \leq x \leq 4$

(OR)

- b. Solve $\frac{\partial^2 u}{\partial x^2} - 2 \frac{\partial u}{\partial t} = 0$, given $u(0, t) = 0$, $u(4, t) = 0$, $u(x, 0) = x(4 - x)$. Assume $h = 1$, find the values of u upto $t = 5$ using Bender-Schmidt recurrence relation.

Reg. No.

B.Tech. DEGREE EXAMINATION, MAY 2019

1st to 7th Semester

15MA206 – NUMERICAL METHODS

(For the candidates admitted during the academic year 2015 – 2016 to 2017-2018)

Note:

- Part - A should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45th minute.
- Part - B and Part - C should be answered in answer booklet.

Time: Three Hours

Max. Marks: 100

PART – A (20 × 1 = 20 Marks)

Answer ALL Questions

- By the method of least squares, the error committed in fitting a straight line is
 (A) $E = \sum y - a \sum xy - b \sum y^2$ (B) $E = \sum y^2 - a \sum xy - b \sum y$
 (C) $E = \sum y - a \sum xy^2 - b \sum y$ (D) $E = \sum y^2 - a \sum x^2 y - b \sum y$
- The order of convergence of Newton's method is
 (A) 2 (B) 3
 (C) 4 (D) 1
- The iteration formula for finding the reciprocal of N by Newton's method is
 (A) $x_{n+1} = x_n(2 - Nx_n)$ (B) $x_{n+1} = x_n(1 - Nx_n)$
 (C) $x_{n+1} = x_n(1 + Nx_n)$ (D) $x_{n+1} = x_n(2 + Nx_n)$
- One of the direct method to solve a system of simultaneous linear equation is
 (A) Gauss Jacobi (B) Gauss Seidal
 (C) Gauss Elimination (D) Newton's Raphson method
- The relation between E and ∇ is
 (A) $\nabla - E^{-1} = 1$ (B) $1 + \nabla = E^{-1}$
 (C) $\nabla = 1 - E^{-1}$ (D) $\nabla = 1 + E^{-1}$
- The relation between E and μ is
 (A) $\mu = E^{1/2} + E^{-1/2}$ (B) $\mu = E^{-1/2} - E^{1/2}$
 (C) $\mu = \frac{1}{2} [E^{1/2} + E^{-1/2}]$ (D) $\mu = \frac{1}{2} [E^{-1/2} - E^{1/2}]$
- When the values of the independent variables are not equally spaced, then we apply
 (A) Newton's forward interpolation formula (B) Central difference interpolation formula
 (C) Newton's backward interpolation formula (D) Lagrange's interpolation formula

8. Extrapolation is the process of
 (A) Finding the values outside the interval (x_0, x_n) (B) Finding the values inside the interval (x_0, x_n)
 (C) Finding the values of the constant (D) Finding the values of the variables
9. The error in the Trapezoidal rule is of order
 (A) $\frac{h^2}{12}$ (B) $\frac{h^3}{6}$
 (C) $\frac{3h^2}{8}$ (D) h^2
10. The truncation error in Simpson's formula is
 (A) $|E| < \frac{(b-a)h^4}{180}$ (B) $|E| < \frac{(b-a)h^3}{180}$
 (C) $|E| < \frac{(b-a)h^4}{12}$ (D) $|E| < \frac{(b-a)h^3}{12}$
11. Trapezoidal rule gives the
 (A) Most accurate result (B) Approximate result
 (C) Least accurate result (D) Simplest result
12. Numerical differentiation is the process of computing the value of
 (A) Δy_0 (B) ∇y_0
 (C) $\frac{\Delta y}{\Delta x}$ (D) $\frac{dy}{dx}$
13. The modified Euler method is based on the average of
 (A) Slopes (B) Curves
 (C) Both slopes and points (D) Points
14. The error term in Milne's corrector formula is
 (A) $\frac{-h}{90} \Delta^4 y_0$ (B) $\frac{h}{90} \Delta^4 y_0$
 (C) $\frac{-h}{90} \Delta^4 y_0$ (D) $\frac{h}{90} \Delta^4 y_0$
15. Predictor corrector methods are
 (A) Single step method (B) Self starting method
 (C) Multistep method (D) Self correcting method
16. Using Taylor's series method, the value $y(1.1)$ given that $y' = x + y, y(1) = 0, h = 0.1$ is
 (A) 0.1103 (B) 1.103
 (C) 0.2 (D) 0.1
17. The nature of the partial differential equation $f_{xy} - f_x = 0$ is
 (A) Elliptic (B) Parabolic
 (C) Hyperbolic (D) Laplace equation

18. $u_{i,j} = \frac{1}{4} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$ is known as
 (A) Diagonal five point formula (B) Standard five point formula
 (C) Crank-Nicholson formula (D) Bender-Schmidt formula
19. Bender-Schmidt scheme converges for
 (A) $\lambda = 1$ (B) $\lambda = 1/2$
 (C) $\lambda = 3$ (D) $\lambda = 3/2$
20. The solution of hyperbolic equation is stable only if λ is
 (A) 1 (B) $1/2$
 (C) a (D) $1/a$

PART – B (5 × 4 = 20 Marks)

Answer ANY FIVE Questions

21. Find a positive root of $xe^x = 2$ by the method of false position.
22. Express $f(x) = x^3 - 3x^2 + 5x + 7$ in terms of factorial polynomial taking $h = 2$ and find its differences.
23. From the following table, find the missing value

x	2	3	4	5	6
$f(x)$	45.0	49.2	54.1	-	67.4

24. Prove that $EV = \nabla = \nabla E$.
25. Find $\sin 52^\circ$ by using Newton's forward interpolation formula for the following table.

y	45°	50°	55°	60°
$y = \sin x$	0.7071	0.7660	0.8192	0.8660

26. Evaluate $\int_0^1 \frac{dx}{1+x^2}$, using Trapezoidal rule with $h = 0.2$.
27. Compute y at $x = 0.25$ by Modified Euler method given $y' = 2xy, y(0) = 1$.

PART – C (5 × 12 = 60 Marks)

Answer ALL Questions

28. a. Fit a parabola, by the method of least squares, to the following data. Also estimate y at $x = 6$.

x	1	2	3	4	5
y	5	12	26	60	97

(OR)

- b. Solve the following system of equations by Gauss Seidal method correct to three decimal places $27x + 6y - z = 85; 6x + 15y + 2z = 72; x + y + 54z = 110$

- b. Using Newton's divided difference formula, find the values $f(2)$, $f(8)$ and $f(15)$, given the following table.

x	4	5	7	10	11	13
$f(x)$	48	100	294	900	1210	2028

30. a. Find the value of $\sec 31^\circ$ from the following data:

θ (in degrees)	31°	32°	33°	34°
$\tan \theta$	0.6008	0.6249	0.6494	0.6745

(OR)

- b. By dividing the range into ten equal parts, evaluate $\int_0^\pi \sin x \, dx$ by Trapezoidal and Simpson's rule. Verify your answer with actual integration.

31. a. Apply the fourth order Runge-Kutta method to find $y(0.1)$ given that $y' = x + y$, $y(0) = 1$.

(OR)

- b. Using Milne's method find $y(4.4)$ given $5xy' + y^2 - 2 = 0$, $y(4) = 1$, $y(4.1) = 1.0049$, $y(4.2) = 1.0097$ and $y(4.3) = 1.0143$.

32. a. Solve $\nabla^2 u = 8x^2y^2$ for square mesh given $u = 0$ on the four boundaries dividing the square into 16 sub-squares of length 1 unit.

(OR)

- b. Solve by Crank-Nicholson method the equation $u_{xx} = u_t$ subject to $u(x, 0) = 0$, $u(0, t) = 0$ and $u(1, t) = t$ for two time steps, taking $h = 1/4$.

Reg. No.

B.Tech. DEGREE EXAMINATION, MAY 2019

3rd to 8th Semester

15MA206 – NUMERICAL METHODS

(For the candidates admitted during the academic year 2015 – 2016 to 2017-2018)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45th minute.
(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: Three Hours

Max. Marks: 100

PART – A (20 × 1 = 20 Marks)

Answer ALL Questions

- Newton-Raphson method is also known as the method of
(A) Secant (B) Normal
(C) Tangents (D) Squares
- The order of convergence of Newton's method is
(A) 2 (B) 3
(C) 4 (D) 1
- The rate of convergence in Gauss-Seidal method is roughly _____ as faster as in Jacobi's method.
(A) One time (B) Two times
(C) Thrice (D) Four times
- Which one of the method is a direct method
(A) False position method (B) Gauss-Seidal method
(C) Gauss-Jacobi method (D) Gauss-Elimination method
- The relation between the operators E and ∇ is
(A) $E = 1 + \nabla$ (B) $\nabla = 1 - E$
(C) $\nabla = 1 - E^{-1}$ (D) $\nabla = 1 + E$
- The n^{th} divided difference of a polynomial is _____.
(A) Zero (B) Constant
(C) n (D) $n + 1$
- The process of finding the value of 'x' given value y is called as
(A) Inverse interpolation (B) Interpolation
(C) Extrapolation (D) Forward and backward interpolation
- The sixth term of the sequence 8, 12, 19, 29, 42
(A) 52 (B) 58
(C) 55 (D) 56
- The error in the Trapezoidal rule is of the order
(A) h (B) h^4
(C) h^2 (D) h^3

10. In Simpson's three-eighth rule $y(x)$ is a polynomial of degree _____.
 (A) 3 (B) 2
 (C) 1 (D) 4
11. Which of the rule is applicable, if the number of intervals n is even?
 (A) Trapezoidal rule (B) Simpson's one-third rule
 (C) Simpson's three-eighth rule (D) Weddle's rule
12. The accuracy of the result in Trapezoidal rule improved by the way of _____.
 (A) Decreasing the number of intervals (B) Increasing the number of intervals
 (C) Increasing the value of h (D) Increasing the average value of h
13. Which one of the method is a multi step method?
 (A) Euler method (B) Taylor series method
 (C) Runge-Kutta method (D) Milne's method
14. The improved Euler method is based on the averages of _____.
 (A) Points (B) Slopes
 (C) Tangents (D) Normal
15. The Runge-Kutta method of second order is same as _____ method.
 (A) Improved Euler (B) Euler
 (C) Adam's Bashforth (D) Modified Euler
16. The Euler's algorithm for solving $\frac{dy}{dx} = f(x, y)$ with initial condition $y(x_0) = y_0$
 (A) $y_{n+1} = y_n - hf(x_n, y_n)$ (B) $y_{n+1} = y_n - f(x_n, y_n)$
 (C) $y_{n+1} = y_n + hf(x_n, y_n); n = 0, 1, 2, \dots$ (D) $y_{n+1} = y_n + f(x_n, y_n)$
17. The partial differential equation $\frac{\partial^2 u}{\partial x^2} + 2\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$ is of _____ type.
 (A) Laplace (B) Elliptic
 (C) Parabolic (D) Hyperbolic
18. An equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$ is called as
 (A) Poisson equation (B) Laplace equation
 (C) Wave equation (D) Heat equation
19. The Bender-Schmidt recurrence equation is
 (A) $u_{i,j+1} = \frac{1}{2}[u_{i,j} + u_{i,j+1}]$ (B) $u_{i,j+1} = \frac{1}{2}[u_{i-1,j} + u_{i+1,j}]$
 (C) $u_{i,j+1} = \frac{1}{2}[u_{i-1,j} - u_{i+1,j}]$ (D) $u_{i,j+1} = \frac{1}{2}[u_{i+1,j} - u_{i,j}]$

20. The Bender-Schmidt recurrence equation is valid only if
 (A) $k = \frac{ah^2}{a}$ (B) $k = ah$
 (C) $k = ah^2$ (D) $k = \frac{ah^2}{4}$

PART - B (5 × 4 = 20 Marks)
 Answer ANY FIVE Questions

21. Find an iterative formula to find $\sqrt{N}$ (where N is a positive number) and hence find $\sqrt{5}$.
22. Express $x^4 + 3x^3 - 5x^2 + 6x - 7$ in factorial polynomial and get their successive forward differences, taking $h = 1$.
23. A curve passes through the points (1, 2), (1.5, 2.4), (2.0, 2.7), (2.5, 2.8) (3, 3), (3.5, 2.6) and (4.0, 2.1). Obtain the area bounded by the curve, the x axis and $x = 1$ and $x = 4$.
24. Solve $\frac{dy}{dx} = x + y$, given $y(1) = 0$, find $y(1.1)$ by Taylor series method.
25. Solve $\frac{\partial^2 u}{\partial x^2} = 2\frac{\partial u}{\partial t}$ given $u(0, t) = 0$, $u(4, t) = 0$, $u(x, 0) = x(4 - x)$. Assume $h = 1$. Find the value of u upto $t = 5$ by Bender-Schmidt method.
26. Find the parabola of the form $y = ax^2 + bx + c$ passing through the points (0, 0), (1, 1) and (2, 20).
27. Form the divided difference table from the following data.

x	1	3	6	11
y	4	32	224	1344

PART - C (5 × 12 = 60 Marks)
 Answer ALL Questions

28. a. Fit a second degree parabola to the following data and also find the error.

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

(OR)

- b. Solve the following system of equations by Gauss Seidal method correct to three decimal places $8x - 3y + 2z = 20$; $4x + 11y - z = 33$; $6x + 3y + 12z = 35$

29. a. The population of a town is as follows.

Years (x)	1941	1951	1961	1971	1981	1991
Population (in lakhs) (y)	20	24	29	36	46	51

Estimate the population increase during the period 1946 to 1976.

(OR)

- b. From the following tables find $f(x)$ and hence $f(4)$ using Newton's formula.

x	-1	1	2	3
y	-21	15	12	3

30. a. A rod is rotating in a plane. The following table gives the angle θ (in radians) through which the rod has turned for various values of time t (seconds). Calculate the angular velocity and angular acceleration of the rod at $t = 0.6$ seconds.

t	0	0.2	0.4	0.6	0.8	1.0
θ	0	0.12	0.49	1.12	2.02	3.20

(OR)

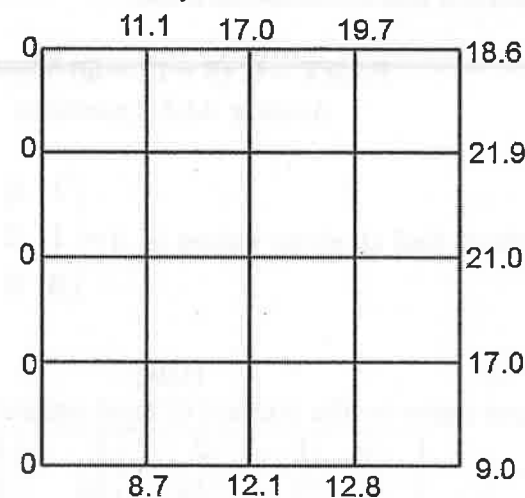
- b. Evaluate $\int_4^{5.2} \log_e x \, dx$ using Trapezoidal and Simpson's rules.

31. a. Using R.K method of fourth order compute $y(0.1), y(0.2)$ given $\frac{dy}{dx} + y + xy^2 = 0, y(0) = 1$, correct to 4 decimals.

(OR)

- b. Find $y(2)$ if $y(x)$ is the solution of $\frac{dy}{dx} = \frac{1}{2}(x + y)$ given $y(0) = 2, y(0.5) = 2.636, y(1) = 3.595, y(1.5) = 4.968$.

32. a. Find the Liebmann's method the values at the interior lattice points of the square region of the harmonic function u whose boundary values are as shown below.



(OR)

- b. Given $f_{xx} = f_t, f(0, t) = f(5, t) = 0, f(x, 0) = x^2(25 - x^2)$, find f in the range taking $h = 1$ and upto 5 seconds.

B.Tech. DEGREE EXAMINATION, NOVEMBER 2018
3rd to 7th Semester

15MA206 – NUMERICAL METHODS

(For the candidates admitted during the academic year 2015 – 2016 to 2017-2018)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45th minute.
(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: Three Hours

Max. Marks: 100

PART – A (20 × 1 = 20 Marks)Answer **ALL** Questions

- The principle of method of least squares state that the best fitting curve to the set of points is that for which the sum of the squares of residuals is
(A) Maximum (B) Minimum
(C) Greater than zero (D) Less than zero
- Newton-Raphson method is also known as the method of _____.
(A) Secant (B) Co-secant
(C) Tangents (D) Squares
- The convergence in the Gauss-Seidel method is roughly _____ as faster as the in Jacobi's method.
(A) Three times (B) Five times
(C) Two times (D) Ten times
- In Gauss-Elimination method for solving a system of linear algebraic equations, the triangularization leads to
(A) Lower triangular matrix (B) Upper triangular matrix
(C) Diagonal matrix (D) Singular matrix
- The $(n+1)^{\text{th}}$ difference of the n^{th} degree polynomial is
(A) Constant (B) Zero
(C) Positive (D) Negative
- $E \nabla =$
(A) E (B) 1
(C) ∇ (D) Δ
- The dependent value y , in the function $y = f(x)$ is known as
(A) Argument (B) Entry
(C) Constant (D) Variable
- $(1 + \Delta)(1 - \nabla) = ?$
(A) 0 (B) 1
(C) δ^2 (D) Δ

9. Simpson's three-eighth rule can be applied only when n is
 (A) Odd (B) Even
 (C) Prime (D) Multiple of 3

10. The error in the Trapezoidal is of order

- (A) $\frac{h^2}{12}$ (B) $\frac{h^3}{6}$
 (C) $\frac{3h^2}{8}$ (D) h^2

11. Numerical differentiation is the process of computing the value of

- (A) Δy_0 (B) ∇y_0
 (C) $\frac{\Delta y}{\Delta x}$ (D) $\frac{dy}{dx}$

12. The truncation error in Simpson's one third rule is

- (A) $|E| < \frac{(b-a)h^2}{12} M$ (B) $|E| < \frac{(b-a)h^4}{180} M$
 (C) $|E| < \frac{(b-a)h^4}{80} M$ (D) $|E| < \frac{(b-a)h^6}{840} M$

13. Taylor's series method is

- (A) Single step method (B) Multi step method
 (C) Iterative method (D) Trial and error method

14. How many prior values are required to predict the next value in Adam's methods

- (A) 1 (B) 2
 (C) 3 (D) 4

15. The error term in Milne's predictor formula is

- (A) $\frac{-19h}{720} \Delta^4 y_0'$ (B) $\frac{14h}{90} \Delta^4 y_0'$
 (C) $\frac{-h}{90} \Delta^4 y_0'$ (D) $\frac{14h}{45} \Delta^4 y_0'$

16. Given $y' = x + y$, $y(0) = 1$, then $y(0.1)$ by Euler's method is

- (A) 0.1 (B) 0.2
 (C) 2.1 (D) 1.1

17. The nature of the PDF $f_{xx} - 2f_{xy} = 0$ is

- (A) Laplace equation (B) Elliptic
 (C) Parabolic (D) Hyperbolic

18. Bender-Schmidt scheme converges for

- (A) $\lambda = 1$ (B) $\lambda = 1/2$
 (C) $\lambda = 3/2$ (D) $\lambda = 3$

19. The error in the diagonal formula is _____ times the error in the standard formula.

- (A) 1 (B) 2
 (C) 3 (D) 4

20. The equation $u_{xx} + u_{yy} = f(x, y)$ is also called

- (A) Laplace equation (B) Poisson's equation
 (C) One dimensional heat equation (D) Two dimensional heat equation

PART - B (5 × 4 = 20 Marks)

Answer ANY FIVE Questions

21. Find the real positive root of $3x - \cos x - 1 = 0$ by the method of tangents.

22. Express $f(x) = x^3 - 3x^2 + 5x + 7$ in terms of factorial polynomial taking $h = 2$ and find its differences.

23. Find the 7th term of the sequence 1, 4, 10, 20, 35, 56, ...

24. Evaluate $\int_{-3}^3 x^4 dx$, by using Trapezoidal rule with $h = 1$.

25. Solve $\frac{dy}{dx} = y^2 + 1$, given $y(0) = 0$ and get $y(0.2)$.

26. Classify the following PDE $(x+1)u_{xx} - 2(x+2)u_{xy} + (x+3)u_{yy} = 0$.

27. State Adam's predictor and corrector formula.

PART - C (5 × 12 = 60 Marks)

Answer ALL Questions

28. a. Using power method, find all eigen values of $A = \begin{pmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

(OR)

- b. Fit a second degree curve by the method of least squares, to the following data.

x	1	2	3	4	5
y	5	12	26	60	97

Also find y at $x = 6$.

29. a. From the data given below, find the number of students whose weight is between 60 and 70lbs.

Weight in lbs	0-40	40-60	60-80	80-100	100-120
No of students	250	120	100	70	50

(OR)

- b. Solve $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$ given $u(0, t) = 0$, $u(4, t) = 0$, $u(x, 0) = x(4 - x)$ assuming $h = k = 1$.

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B.Tech. DEGREE EXAMINATION, JANUARY 2023
Fourth Semester

MA0202 – NUMERICAL METHODS
(For the candidates admitted from the academic year 2007-2008 to 2012-2013)

Time: Three hours

Max. Marks: 100

Answer ALL Questions

PART – A (10 × 2 = 20 Marks)

1. Write the normal equations to fit a curve of the form $y = ae^{bx}$.
2. Write down the Newton-Raphsons formula to find $\sqrt[3]{N}$. N is a positive integer.
3. Construct the divided difference table for the following

x	0	1	2	5
$f(x)$	2	3	12	25

4. Show that $E = (1 - \nabla)^{-1}$.
5. Describe numerical Integration.
6. Evaluate $\int_0^2 e^x dx$ using Trapezoidal rule by taking four intervals.
7. State improved Euler's algorithm for solving $\frac{dy}{dx} = f(x, y)$.
8. Write down Adam-Bashforts predictor and corrector formula.
9. Classify $u_{xx} + 2u_{xy} + u_{yy} = 0$.
10. Write the standard five point formula.

PART – B (5 × 16 = 80 Marks)

- 11.a. Fit a curve of the form $y = ax^b$ to the data given below using method of least squares

x	1	2	3	4	5	6
y	1200	900	600	200	110	50

(OR)

- b. Solve the following system of equations by Gauss Seidel method correct to three decimal places if
 $8x - 3y - 2z = 20$, $4x + 11y - z = 33$, $6x + 3y + 12z = 35$

- 12.a. From the following data, find the value of y at $x = 46$ and $x = 63$.

x	1941	1951	1961	1971	1981	1991
y	20	24	29	36	46	51

(OR)

- b.i. Using Lagrange's formula of interpolation find $y(9.5)$ given

x	7	8	9	10
y	3	1	1	9

and hence find y when $x = 1$.

(10 Marks)

- ii. Express $x^4 + 3x^3 - 5x^2 + 6x - 7$ in factorial polynomials and get their successive forward differences by taking $h=1$. (6 Marks)

- 13.a. Find the first two derivatives of $x^{1/3}$ at $x = 50$ and $x = 56$ given the table below

(x)	50	51	52	53	54	55	56
$y = x^{1/3}$	3.684	3.7084	3.7325	3.7563	3.7798	3.8030	3.8259

(OR)

- b. Evaluate $I = \int_0^6 \frac{1}{1+x} dx$ using

- Trapezoidal rule
- Simpson's rule
- Simpson's $3/8^{\text{th}}$ rule

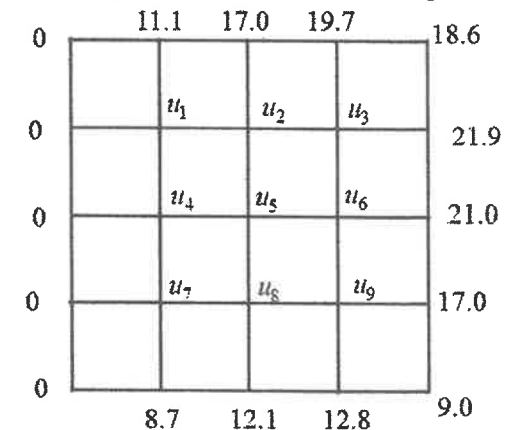
Check up by direct integration.

- 14.a. Given $y'' + xy' + y = 0$, $y(0) = 1$, $y'(0) = 0$, find the values of $y(0.1)$, $y(0.2)$ by using Runge Kutta method of fourth order.

(OR)

- b. Find $y(2)$ if $y(x)$ is the solution of $\frac{dy}{dx} = \frac{1}{2}(x+y)$ given $y(0) = 2$, $y(0.5) = 2.636$, $y(1) = 3.595$ and $y(1.5) = 4.968$.

- 15.a. Determine by iteration method the values at the interior lattice points of a square region of the harmonic function u whose boundary values are given as shown in the figure below:



(OR)

30. a. The table given below reveals the velocity v of a body during the time ' t ' specified. Find its acceleration at $t = 1.1$.

t	1	1.1	1.2	1.3	1.4
v	43.1	47.7	52.1	56.4	60.8

(OR)

- b. Evaluate $I = \int_0^6 \frac{1}{1+x} dx$ using (i) Trapezoidal rule (ii) Simpson's $1/3^{\text{rd}}$ rule and (iii) Simpson's $3/8$ rule.
31. a. Apply the fourth order Runge-Kutta method to find $y(0.1)$ and $y(0.2)$ given that $y' = x + y, y(0) = 1$.

(OR)

- b. Find $y(2)$ if $y(x)$ is the solution of $\frac{dy}{dx} = \frac{x+y}{2}$ given $y(0) = 2, y(0.5) = 2.636, y(1) = 3.595$ and $y(1.5) = 4.968$.
32. a. Find the Liebmann's method the values of the interior lattice points of a square region of the harmonic function u whose boundary values are as shown below.

	11.1	17	19.7	18.6
0				
0				21.9
0				21.0
0				17.0
0				9.0
	8.7	12.1	12.8	

(OR)

- b. Solve $\frac{\partial^2 u}{\partial x^2} - 2 \frac{\partial u}{\partial t} = 0$ given $u(0, t) = 0, u(4, t) = 0, u(x, 0) = x(4 - x)$. Assume $h = 1$ find the values of u upto $t = 5$.

Reg. No.

B.Tech. DEGREE EXAMINATION, NOVEMBER 2018
Fifth Semester

16

MA1025 – NUMERICAL METHODS AND ITS APPLICATIONS
(For the candidates admitted during the academic year 2013 – 2014 and 2014 -2015)

Note:

- (i) Part - A should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45th minute.
- (ii) Part - B and Part - C should be answered in answer booklet.

Time: Three Hours

Max. Marks: 100

PART – A (20 × 1 = 20 Marks)
Answer ALL Questions:

- The principle of method of least squares states that the best fitting curve to the set of points is that for which the sum of squares of residual is
(A) Minimum (B) Maximum
(C) Greater than zero (D) Less than zero
- The condition for convergence of a function $f(x)$ in Newton-Raphson method
(A) $|f'(x)f''(x)| > |f(x)|^2$ (B) $|f(x)f'(x)| < |f''(x)|^2$
(C) $|f(x)f''(x)| < |f'(x)|^2$ (D) $|f(x)f''(x)| > |f'(x)|^2$
- One of the direct method to solve a system of simultaneous linear equation is
(A) Gauss-Jacobi (B) Gauss-Elimination
(C) Gauss-Seidel (D) Newton's method
- The order of convergence of Newton's method is
(A) 2 (B) 3
(C) 4 (D) 1
- The independent variable x , in the function $y = f(x)$ is called as
(A) Variable entry (B) Entry
(C) Argument (D) Constant
- The translation operator E is defined by
(A) $Ef(x) = f(x+h)$ (B) $Ef(x) = f(x)$
(C) $Ef(x+h) = f(x)$ (D) $Ef(x) = f(x-h)$
- The relation between E and μ is
(A) $\mu = E^{1/2} + E^{-1/2}$ (B) $\mu = E^{1/2} - E^{-1/2}$
(C) $\mu = \frac{1}{2} [E^{1/2} + E^{-1/2}]$ (D) $\mu = \frac{1}{2} [E^{-1/2} - E^{1/2}]$
- The n^{th} forward difference of y are denoted by
(A) $\Delta y_{n-1} = y_n - y_{n-1}$ (B) $\Delta y_n = y_{n-1} + y_{n+1}$
(C) $\Delta y_{n-1} = y_n + y_{n-1}$ (D) $\Delta y_n = y_{n-1} + y_n$

9. The error in the Trapezoidal is of order
 (A) $h^2/12$ (B) $h^3/6$
 (C) $3h^2/8$ (D) h^2
10. Simpson's three-eighth rule which can be applied when
 (A) Number of ordinates is odd (B) Number of ordinates is even
 (C) 'n' is multiple of three (D) h is even
11. Newton's backward difference formula to get the derivative of y(x) at any x is
 (A) $\frac{dy}{dx} = \frac{1}{h} \left[\nabla y_n + \left(\frac{2v+1}{2} \right) \nabla^2 y_n + \dots \right]$ (B) $\frac{dy}{dx} = \frac{1}{h} \left[\nabla y_0 + \left(\frac{2u-1}{2} \right) \nabla^2 y_0 + \dots \right]$
 (C) $\frac{dy}{dx} = \frac{1}{h} \left[\Delta y_n + \left(\frac{2v+1}{2} \right) \nabla^2 y_n + \dots \right]$ (D) $\frac{dy}{dx} = \frac{1}{h} \left[\nabla y_n + \left(\frac{v+1}{2} \right) \nabla^2 y_n + \dots \right]$
12. Trapezoidal rule gives the
 (A) Most accurate result (B) Approximate result
 (C) Least accurate result (D) Simplest result
13. Taylor's series method is
 (A) Single step method (B) Multi step method
 (C) Iterative method (D) Trial and error method
14. Given $y' = x + y$, $y(0) = 1$, then $y(0.1)$ by Euler's method, taking $h = 0.1$ is
 (A) 1 (B) 0.2
 (C) 0.1 (D) 1.1
15. Which of the following is better method
 (A) Taylor series method (B) Euler's method
 (C) R-K method (D) Modified Euler method
16. Predictor corrector methods are
 (A) Single step method (B) Self starting method
 (C) Multi step method (D) Self correcting method
17. If $B^2 - 4AC < 0$, then the second order PDE is known as
 (A) Elliptic (B) Parabolic
 (C) Hyperbolic (D) One dimensional wave equation
18. $\frac{\partial^2 u}{\partial x^2} = \frac{1}{\alpha^2} \frac{\partial u}{\partial t}$ is called
 (A) Elliptic (B) Parabolic
 (C) Hyperbolic (D) Laplace equation
19. Bender-Schmidt scheme converges for
 (A) $\lambda = 1$ (B) $\lambda = 1/2$
 (C) $\lambda = 3$ (D) $\lambda = 3/2$

20. $u_{i,j+1} = \frac{1}{4} [u_{i-1,j+1} + u_{i+1,j+1} + u_{i-1,j} + u_{i+1,j}]$ is known as
 (A) Standard five point formula (B) Diagonal five point formula
 (C) Bender-Schmidt formula (D) Crank-Nicholson formula

PART - B (5 × 4 = 20 Marks)
 Answer ANY FIVE Questions

21. Find an iterative formula to find the reciprocal of a given number N and hence find the value of $\frac{1}{19}$.
22. Express $f(x) = x^3 - 3x^2 + 5x + 7$ in terms of factorial polynomial taking $h = 2$ and find its differences.
23. From the following table, find the missing value.
- | | | | | | |
|------|----|------|------|---|------|
| x | 2 | 3 | 4 | 5 | 6 |
| f(x) | 45 | 49.2 | 54.1 | - | 67.4 |
24. Write $\frac{dy}{dx}$ at $x = x_0$ and $x = x_n$ by using Newton's forward and backward differences.
25. Using Euler's method, solve numerically the equation $y' = x + y$, $y(0) = 1$, for $x = 0.2$ and 0.4 .
26. Write the Adam's predictor and corrector formula.
27. Classify the following equation $(x+1)u_{xx} - 2(x+2)u_{xy} + (x+3)u_{yy} = 0$.

PART - C (5 × 12 = 60 Marks)
 Answer ALL Questions

28. a. Fit a straight line and a parabola to the following data.

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

(OR)

- b. Solve the following system of equations by using Gauss-Seidel method (Correct to 3 decimal place) $8x - 3y + z = 20$; $4x + 11y - z = 33$; $6x + 3y + 12z = 35$.

29. a. From the following data, find θ at $x = 43$ and $x = 84$.

x	40	50	60	70	80	90
θ	184	204	226	250	276	304

(OR)

- b. Using Lagrange's interpolation formula, find $y(0)$ from the following table.

x	5	6	9	11
y	12	13	14	16